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3.1.3 Production, Costs and Revenue

Production and Productivity

Production is the process of converting inputs (factors of production — land, labour, capital, enterprise) into outputs (goods and services).

Productivity is a measure of efficiency — the output produced per unit of input in a given time period. Higher productivity means more output from the same resources.

Measuring Productivity

$$\text{Labour productivity} = \text{Total output} \div \text{Number of workers (or labour hours)}$$

Productivity can be increased through:

- **Better technology:** new machinery, automation, AI tools
- **Education and training:** improving human capital and worker skills
- **Better management:** more efficient work organisation and processes
- **Specialisation:** workers focusing on tasks where they have expertise
- **Capital investment:** providing workers with better tools and equipment

Real-World Example: UK Productivity Puzzle

Since the 2008 financial crisis, UK productivity growth has stagnated — a phenomenon economists call the "productivity puzzle." Between 1970 and 2007, output per hour grew by about 2% per year. Since 2008, it has barely grown at all. Possible explanations include: underinvestment in capital, a shift towards low-productivity service jobs, zombie firms surviving on low interest rates, and slower

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technology adoption. This matters because productivity growth is the main driver of rising living standards in the long run.

The Short Run and the Long Run

The short run is the time period in which at least one factor of production is *fixed* (cannot be changed). Typically, capital is fixed in the short run — a firm cannot instantly build a new factory. Only variable factors (like labour and raw materials) can be adjusted.

The long run is the time period in which *all* factors of production are variable. The firm can change its entire scale of production — build new factories, buy new machinery, enter or exit the industry entirely.

Important: the short run and long run are not specific time periods (not "6 months" or "5 years"). They are defined by the fixity of factors and vary by industry. For a software firm, the long run might be a few weeks (just hire more developers). For a nuclear power station, the long run might be 10+ years (building a new plant takes that long).

Exam Tip: Always be clear about whether you are discussing short-run or long-run analysis. The cost curves, the behaviour of firms, and the market outcomes are different in each time period. Mixing them up is a common source of lost marks.

The Law of Diminishing Marginal Returns

The Law of Diminishing Marginal Returns states that as additional units of a variable factor (e.g., labour) are added to one or more fixed factors (e.g., capital/machinery), there comes a point beyond which the marginal product (the extra output from one more unit of the variable factor) begins to **fall**.

This is a **short-run concept only** — it requires at least one factor to be fixed. It does NOT apply in the long run when all factors can be changed.

Detailed Example: A Coffee Shop

Imagine a coffee shop with 2 espresso machines (fixed capital), a fixed amount of counter space, and a fixed number of tables:

Number of Workers	Total Output (coffees/hour)	Marginal Product (MP)	Stage
0	0	—	
1	15	15	Increasing marginal returns — workers specialise (one takes orders, one makes coffee)
2	35	20	
3	60	25	
4	80	20	Diminishing marginal returns — machines are being shared, workers start getting in each other's way
5	95	15	
6	105	10	
7	108	3	Severely diminishing returns — far too many workers for the space and equipment
8	105	-3	Negative returns — workers are so crowded they reduce total output

Why This Law Matters for Costs

Diminishing marginal returns directly explain why short-run cost curves are U-shaped:

- When marginal product is **rising** (increasing returns): each extra worker produces more output, so the cost of each extra unit of output is **falling** → marginal cost falls
- When marginal product is **falling** (diminishing returns): each extra worker produces less output, so the cost of each extra unit of output is **rising** → marginal cost rises

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This is why MC, AVC, and AC are all U-shaped — they first fall (increasing returns phase) then rise (diminishing returns phase).

Exam Tip: Do NOT confuse the law of diminishing marginal *returns* (short run, at least one fixed factor) with decreasing returns to *scale* (long run, all factors variable). This is one of the most common errors in A-Level Economics and will cost you marks every time.

Short-Run Costs

Types of Cost

Fixed costs (FC) — costs that do *not* vary with the level of output in the short run. They must be paid even if output is zero. Examples: rent, insurance premiums, salaries of permanent staff, depreciation of machinery, business rates.

Variable costs (VC) — costs that *do* vary directly with the level of output. If output is zero, variable costs are zero. Examples: raw materials, energy/electricity for production, hourly wages, packaging.

Total cost (TC) = Fixed costs + Variable costs (TC = FC + VC)

Average and Marginal Costs

Cost Measure	Formula	What It Tells You	Shape
Average Fixed Cost (AFC)	FC ÷ Q	Fixed cost per unit of output. Falls continuously as output rises — the fixed cost is "spread" over more units.	Downward-sloping (asymptotic to x-axis)
Average Variable Cost (AVC)	VC ÷ Q	Variable cost per unit. Falls initially (increasing returns), then rises (diminishing returns).	U-shaped

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Average Total Cost (AC or ATC)	TC ÷ Q = AFC + AVC	Total cost per unit. The gap between AC and AVC equals AFC (which narrows as output rises).	U-shaped
Marginal Cost (MC)	ΔTC ÷ ΔQ	The additional cost of producing one more unit of output. The most important cost curve for decision-making.	U-shaped (falls then rises sharply)

Short-Run Cost Curves Diagram

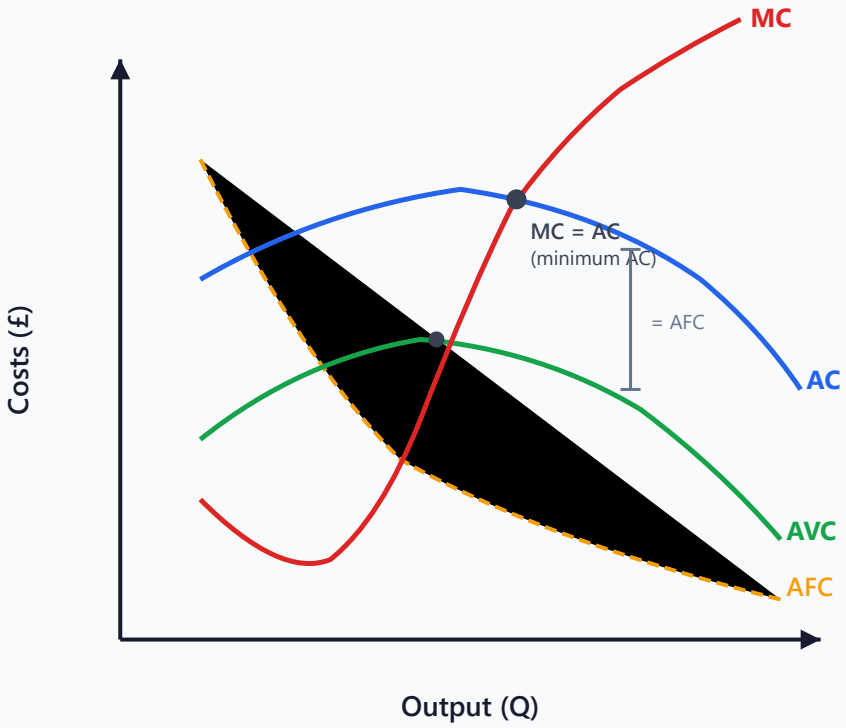

Figure 1: Short-run cost curves — MC crosses both AVC and AC at their minimum points. The gap between AC and AVC equals AFC.

Critical Relationships Between Cost Curves

These relationships are essential for exam diagrams — get them right every time:

1. **MC crosses AC at the minimum point of AC.** When MC < AC, average cost is falling (being pulled down). When MC > AC, average cost is rising (being pulled up). Think of it like exam grades: if your

next grade (marginal) is below your average, it pulls your average down.

2. **MC crosses AVC at the minimum point of AVC** — same logic applies.
3. **AFC falls continuously** — it never turns upward because fixed costs are constant being divided by ever-larger output.
4. **The vertical gap between AC and AVC equals AFC** — this gap narrows as output increases (because AFC falls).
5. **The U-shape of all curves is caused by diminishing marginal returns.**

> **Real-World Example: Why Airlines Sell Cheap Last-Minute Seats**
> Airlines have very high fixed costs (aircraft leases, airport fees, crew salaries) and relatively low variable costs per additional passenger (a bit of fuel, a meal). Once a flight is scheduled, the marginal cost of filling an empty seat is very low — perhaps £20-30. This is why airlines sell remaining seats cheaply before departure. As long as the price covers the marginal cost, the airline is better off selling the seat than flying empty. This is the MC pricing principle in action.

Long-Run Costs

In the long run, **all** factors of production are variable. There are no fixed costs. The firm can choose its optimal scale — the size of factory, amount of machinery, number of permanent staff. The key concept is the **Long-Run Average Cost (LRAC)** curve.

The LRAC Curve

The LRAC curve shows the lowest possible average cost of producing each level of output when the firm can vary all factors. It is derived as an "envelope" of many short-run average cost (SRAC) curves — each SRAC represents a different plant size. The firm chooses whichever plant size minimises cost for its desired output level.

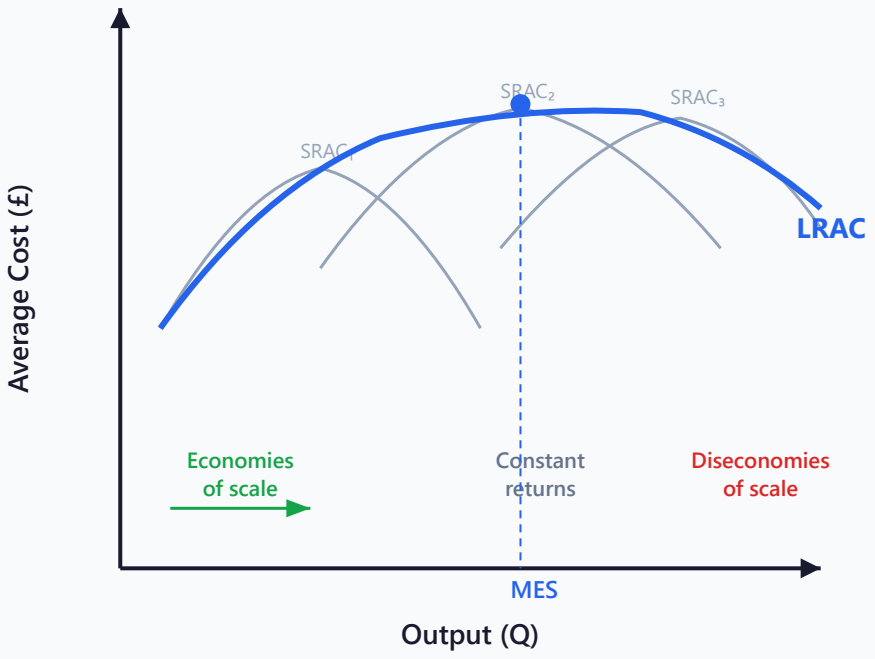

Figure 2: The LRAC curve showing economies of scale, constant returns, diseconomies of scale, and the Minimum Efficient Scale (MES)

Economies of Scale

Economies of scale are the cost advantages that a firm gains as it increases its scale of production in the long run. As output rises, long-run average costs **fall**. They explain the downward-sloping section of the LRAC curve.

Internal Economies of Scale

These arise from the growth of the firm itself:

Type	Explanation	Real-World Example
Technical	Larger firms can use more efficient, specialised machinery and production processes. The *container principle*: doubling	Amazon's massive automated warehouses process millions of items at a fraction of the cost

	the surface area of a container more than doubles its volume — so larger containers, tankers, and warehouses are more cost-effective per unit.	per item compared to a small shop's stockroom
Managerial	Larger firms can afford to hire specialist managers (marketing director, head of finance, HR director), each expert in their field. Small firms often have one person doing everything less efficiently.	A multinational like Unilever has entire teams dedicated to supply chain optimisation — a corner shop owner does it all themselves
Purchasing (bulk-buying)	Larger firms buy raw materials in huge volumes and can negotiate significant discounts from suppliers. Their bargaining power reduces input costs per unit.	Tesco buys milk by the millions of litres and negotiates much lower prices than a small convenience store can
Financial	Larger firms can borrow money at lower interest rates because banks see them as lower risk. They also have access to a wider range of financing (bonds, share issues).	Apple can borrow at near-government rates; a local startup might pay 8-12% on a bank loan
Marketing	The cost of advertising is spread over a much larger output. A £1 million TV advert is expensive for a firm selling 10,000 units (£100 per unit) but cheap for a firm selling 10 million (£0.10 per unit).	Coca-Cola's global advertising budget is huge in total but tiny per bottle sold
Risk-bearing	Large firms can diversify across products, markets, and countries. If one product fails, others compensate. This reduces overall business risk.	Samsung makes phones, TVs, semiconductors, washing machines, and ships — a failure in one area doesn't threaten the whole company

External Economies of Scale

These arise from the growth of the *industry* rather than the individual firm:

- **Skilled labour pool:** As an industry grows in an area, a concentration of skilled workers develops. Firms benefit from easy access to trained workers without having to train them from scratch. Example: Silicon Valley for tech workers, the City of London for financial professionals.
- **Supplier networks:** Component suppliers and specialist services cluster near large industries, reducing transport costs and delivery times. Example: car parts suppliers cluster around car manufacturing plants in the Midlands.
- **Knowledge spillovers:** Firms in the same area share ideas, innovations, and best practices — through worker mobility, conferences, and informal networks. Example: biotech firms clustered around Cambridge benefit from proximity to university research.
- **Infrastructure:** Governments invest in transport, broadband, and facilities in areas where industries concentrate. Example: improved road networks near industrial parks.

Real-World Example: Amazon's Economies of Scale

Amazon is a textbook example of economies of scale in action. Its massive automated warehouses (technical economies), specialist logistics teams (managerial), bulk purchasing power over suppliers (purchasing), ability to borrow cheaply (financial), and global advertising spread across billions of orders (marketing) all give it dramatically lower average costs than smaller retailers. Amazon's fulfilment cost per unit has fallen consistently as it has scaled, allowing it to undercut competitors on price while still generating profit.

Diseconomies of Scale

Diseconomies of scale occur when a firm becomes so large that long-run average costs begin to **rise** as output increases. The firm has grown beyond its optimal size. This explains the upward-sloping section of the LRAC curve.

Diseconomies are primarily caused by **management and organisational problems**:

Cause	Explanation	Example
Communication breakdown	In very large organisations, information must pass through many layers of	A directive from the CEO of a 100,000-employee firm may be

	management. Messages get distorted, delayed, or lost entirely.	misinterpreted by the time it reaches the shop floor
Coordination difficulties	As firms grow, coordinating activities across departments, divisions, and countries becomes increasingly complex and expensive.	General Motors in the 2000s had overlapping car brands (Chevrolet, Pontiac, Saturn, Buick) that competed with each other internally
Worker alienation	In very large firms, workers may feel like insignificant "cogs in a machine." This reduces motivation, job satisfaction, and productivity, increasing costs.	High turnover rates at Amazon warehouses have been linked to worker dissatisfaction, increasing recruitment and training costs
Bureaucracy	Excessive layers of management and complex approval processes slow decision-making. The firm becomes inflexible and unresponsive to market changes.	Nokia's bureaucratic structure meant it was slow to respond to the smartphone revolution, losing market share to Apple and Samsung
Principal-agent problem	As firms grow, it becomes harder for owners (principals) to monitor and control managers (agents), who may pursue their own interests rather than profit maximisation.	Managers may expand their departments for prestige even when it doesn't benefit the company

Minimum Efficient Scale (MES)

Minimum Efficient Scale (MES) is the lowest level of output at which all internal economies of scale have been fully exploited and LRAC is at its minimum. Beyond MES, the firm cannot reduce average costs further by growing.

MES and Market Structure

The relationship between MES and total market demand determines the likely structure of the market:

MES Relative to Market	Market Structure	Example
MES is very large relative to market demand	**Natural monopoly** — only one firm can operate efficiently	Water supply, electricity grid, railways
MES is large relative to market demand	**Oligopoly** — only a few firms can reach efficient scale	Car manufacturing, aircraft (Boeing & Airbus), supermarkets
MES is small relative to market demand	**Competitive market** — many firms can coexist at efficient scale	Restaurants, hairdressers, plumbers, small retail

Returns to Scale (Long Run)

Returns to scale describe what happens to output when **all inputs are increased proportionally** in the long run:

Type	What Happens	Implication for LRAC
Increasing returns to scale	Output increases by *more* than the proportional increase in inputs. E.g., doubling all inputs more than doubles output.	LRAC is falling (economies of scale)
Constant returns to scale	Output increases by exactly the same proportion as inputs. E.g., doubling all inputs exactly doubles output.	LRAC is constant (flat section)
Decreasing returns to scale	Output increases by *less* than the proportional increase in inputs. E.g., doubling all inputs less than doubles output.	LRAC is rising (diseconomies of scale)

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Exam Tip — CRITICAL DISTINCTION:
Diminishing marginal returns = SHORT RUN. One factor is fixed, the other varies. It explains why short-run cost curves are U-shaped.
Decreasing returns to scale = LONG RUN. ALL factors are variable and are increased together. It explains why LRAC eventually rises.
These are completely different concepts. Confusing them is one of the most penalised mistakes in A-Level Economics.

Revenue

Total Revenue (TR) = Price × Quantity sold (TR = P × Q)
Average Revenue (AR) = TR ÷ Q = Price (in a single-price market, AR is always equal to price)
Marginal Revenue (MR) = the change in total revenue from selling one additional unit (MR = ΔTR ÷ ΔQ)

Revenue Under Perfect Competition

In a perfectly competitive market, the firm is a **price taker** — it can sell any quantity at the market price but cannot influence that price. Therefore:

- AR = MR = Price (all constant, represented by a horizontal line)
- TR increases linearly with output (straight line from origin)
- The firm's demand curve is perfectly elastic (horizontal)

Revenue Under Imperfect Competition (Monopoly/Oligopoly)

When a firm has some market power, it faces a **downward-sloping demand curve**. To sell more units, it must lower its price — and crucially, it must lower the price on *all* units, not just the extra one. This has important consequences:

- **AR (= demand curve)** slopes downward
- **MR lies below AR** — because to sell one more unit, the firm must reduce the price on all previous units too. The marginal revenue from an extra sale is less than the price received for it.

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- For a **linear demand curve**, MR has the same intercept but **falls twice as steeply** as AR
- **TR is maximised where MR = 0** — beyond this point, selling more actually reduces total revenue

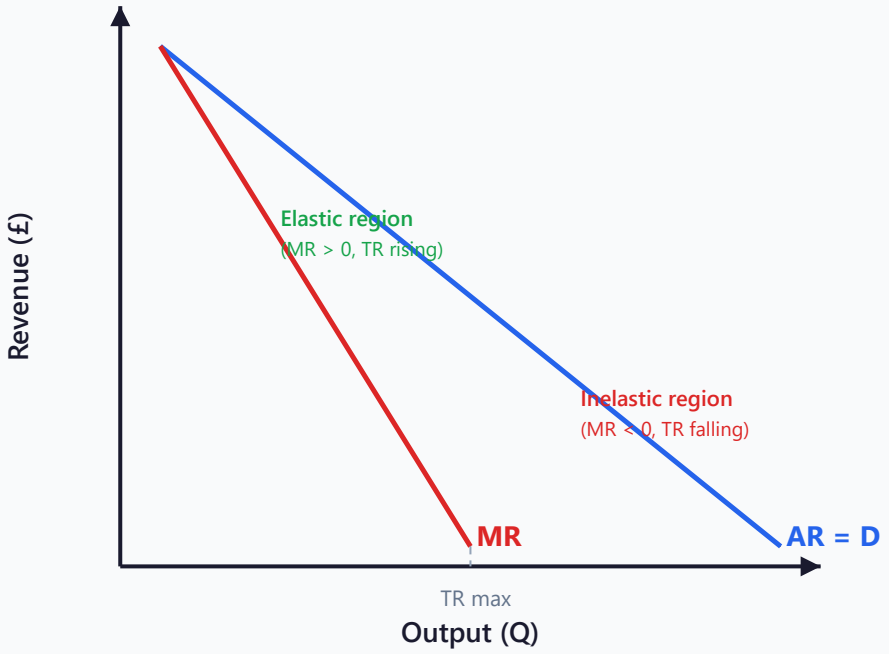

Figure 3: AR and MR for a firm with market power. MR falls twice as fast as AR. TR is maximised where MR = 0.

Profit

Profit = Total Revenue − Total Cost (TR − TC)

Normal profit: The minimum level of profit needed to keep a firm in the industry in the long run. It is the opportunity cost of the entrepreneur's time and capital — what they could earn in their next best alternative. At normal profit, TR = TC (because normal profit is included as a cost).

Supernormal (abnormal/economic) profit: Profit above normal profit. TR > TC. This is a signal that attracts new firms into the market.

Subnormal profit (economic loss): Profit below normal profit. TR < TC. This signals that resources are better used elsewhere and encourages firms to exit.

Profit Maximisation: MC = MR

The standard assumption is that firms maximise profit. A firm maximises profit by producing the quantity where **MC = MR** (and MC is rising through MR). The logic:

- For any unit where **MR > MC**: producing that unit adds more to revenue than to cost → it adds to profit → produce it
- For any unit where **MR < MC**: producing that unit adds more to cost than to revenue → it reduces profit → don't produce it
- Therefore, profit is maximised at the exact point where **MR = MC**

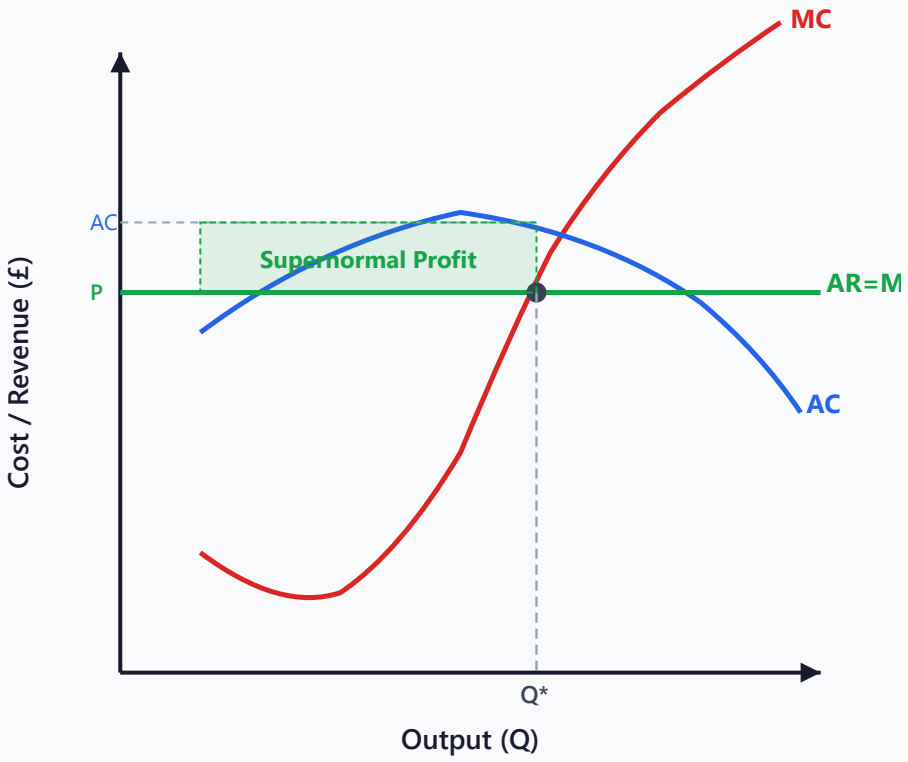

*Figure 4: Profit maximisation at MC = MR. Supernormal profit (green area) = (P − AC) × Q**

The Shut-Down Condition

Even loss-making firms may continue producing in the short run:

- **Short run:** A firm should continue producing as long as **AR ≥ AVC**. If the firm shuts down, it still has to pay fixed costs. As long as revenue covers variable costs and contributes something towards fixed costs, the firm loses less by continuing than by shutting down. If **AR < AVC**, the firm should shut down immediately — it's losing money on every unit produced.
- **Long run:** A firm should exit the industry if **AR < AC** (it cannot cover all costs, including fixed costs that are now variable). There's no point staying in business if you can't earn at least normal profit.

Alternative Objectives of Firms

The assumption that all firms maximise profit (MC = MR) is a simplification. In reality, firms pursue a range of objectives depending on their ownership structure, competitive environment, and managerial incentives:

Objective	Decision Rule	Why a Firm Might Choose This	Example
Profit maximisation	MC = MR	Traditional assumption. Owners want maximum returns on investment.	Small owner-managed businesses
Revenue maximisation	MR = 0	Managers' bonuses may be linked to sales revenue rather than profit. Maximising revenue means selling more than profit max.	Sales-driven tech companies in growth phase
Sales maximisation	AC = AR (normal profit constraint)	Maximise output/market share while still covering costs. Dominance in the market can have long-term strategic benefits.	Amazon sacrificed profits for years to maximise market share
Satisficing	Achieve "satisfactory" levels of profit, not maximum	In large firms, managers (agents) may not share owners' (principals') goal of maximum profit. They balance multiple stakeholder interests — enough profit to keep shareholders happy, enough	Large PLCs where ownership is separated from control

		wages to keep workers, enough investment to keep the firm competitive.	
Survival	Cover costs / stay solvent	New firms or firms in a recession focus on staying alive rather than maximising profit.	Startups, firms during COVID-19 lockdowns
Social/ethical objectives	Consider wider stakeholder impacts	Some firms prioritise environmental sustainability, employee welfare, or community impact alongside profit.	Patagonia, The Body Shop, social enterprises

Evaluation: Is Profit Maximisation Realistic?

- **Information problem:** Firms rarely know their exact MC and MR curves. They use rules of thumb (cost-plus pricing) rather than calculating MC = MR precisely.
- **Principal-agent problem:** In large PLCs, managers run the firm day-to-day but may pursue their own interests (higher salary, empire-building, easy life) rather than maximising profit for shareholders.
- **Stakeholder pressure:** Firms face pressure from employees, customers, governments, and the public — not just shareholders.
- **Short-run vs long-run:** A firm might sacrifice short-run profit (e.g., predatory pricing) to achieve long-run dominance.
- **However:** In competitive markets, firms that don't approximate profit maximisation will be driven out by those that do. Profit maximisation remains the best single assumption for predicting firm behaviour.

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